

# Gell-Mann–Nishijima from the Galois Structure

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4 September 2026

## Abstract

The Gell-Mann–Nishijima relation  $Q = I_3 + Y/2$  is derived from two established Galois properties: (1) Hypercharge  $Y = -1 + \frac{4}{3}\text{NQR}_{13}$  from the Legendre symbol mod 13 (Doc. 346, Theorem A). (2) Isospin  $I_3 = \pm\frac{1}{2}$  from the Su/Sd split via the jE7 projector (Doc. 344, Theorem E). The combination gives all four fermion charges  $\{0, -1, +\frac{2}{3}, -\frac{1}{3}\}$  exactly. Theorem E [S] of Doc. 346 is closed: **[B]**. See German version for full proofs.

## .1 What remains open: $\text{SU}(2)_L$ and CKM [S]

**$\text{SU}(2)_L$  chirality.** Chirality requires  $\gamma_5 = e_1 \cdots e_6 \in \text{Cl}(6)$ . The  $\mathbb{Z}_2$  parity in  $\text{GF}(27)^*$  separates particles from anti-particles ( $k < 13$  vs.  $k \geq 13$ ), not left-handed from right-handed. All four primitive orbits  $O_1 \dots O_4$  are same-chiral. What is needed: the bridge  $\text{GF}(27)^* \rightarrow \text{Cl}(6) \xrightarrow{\gamma_5}$  chirality. The embedding of Galois orbits into the Clifford grade structure is begun in Docs. 341/344 but not yet complete. **[S]**

**CKM mixing angles.**  $\text{GF}(27)$  carries a trace bilinear form  $\langle a, b \rangle = \text{Tr}_{\text{GF}(27)/\text{GF}(3)}(ab) = ab + (ab)^3 + (ab)^9$ . The Cabibbo angle between generations would be the overlap  $|\langle O_{2,\text{Gen.1}}, O_{4,\text{Gen.2}} \rangle| / (\|O_2\| \|O_4\|)$ . Experimental:  $\sin \theta_C \approx 0.225$ ; Galois candidate  $\sin(\pi/14) \approx 0.222$  (1.2 %). Full derivation requires: (1) explicit trace form computation for all orbit pairs; (2) generation assignment within orbits; (3) CKM matrix evaluation. A concrete, executable next step requiring its own document. **[S]**

# Bibliography

- [1] J. Pascher, *Doc. 344*, FFGFT corpus (Sept. 2026).
- [2] J. Pascher, *Doc. 346*, FFGFT corpus (Sept. 2026).